Analysis of the Recovery Rate and Separation Quality Dip at Narrow Generation Angles in the Hit-Competition Study

`hit_competition_study.ipynb` — Toy Characterisation Series

G. Scriven

Quantum Track Reconstruction Study
LHCb VELO Toy Model

April 2026

Contents

Notation and Symbols

The following tables collect every symbol used in this report, grouped by category.

Detector Geometry

Symbol	Definition
M	Number of detector modules (= 5)
Δz	Longitudinal spacing between consecutive modules (= 33 mm)
z_k	z-position of module k (z_1=100, z_2=133, z_3=166, z_4=199, z_5=232 mm)
z_0	z-position of the primary vertex (= 50 mm)
$L_{x,y}$	Detector half-size in the transverse plane (= 50 mm)

Physics and Generation Parameters

Symbol	Definition
n	Number of tracks per event
ϕ_{max}, θ_{max}	Half-angle of the generation cone in ϕ and θ
A_{cone}	Angular area of the generation cone = $(2\phi_{\text{max}})^2$
N_{rep}	Number of independent repeat events per density point
σ_{res}	Hit position resolution (= 0 mm in this study)
σ_{scatt}	Multiple-scattering angular width (= 10^{-4} rad)
k	Acceptance scale factor for the angular tolerance (= 3.0)
θ_{min}	Minimum angular offset (= 1.5×10^{-5} rad)
θ_r	Angular resolution term = $\arctan(k\,\sigma_{\text{res}}/\Delta z)$

Hamiltonian Model Parameters

Symbol	Definition
ε	Angular acceptance tolerance for segment coupling (= 0.425 mrad); defined by Eq. (1)
γ	Self-interaction (diagonal penalty) weight (= 1.5)
δ	Bias term (= 1.0)
β	Baseline activation = $\delta/(\delta + \gamma)$ (= 0.400)
τ	Decision threshold = $(1 + \beta)/2$ (= 0.700)

Segment and Matrix Quantities

Symbol	Definition
N_{seg}	Total number of candidate segments $= (M-1)\,n^2$
$\mathbf{A}$	Hamiltonian system matrix (stored form: positive diagonal, negative off-diagonal)
A_{ij}	Entry of the system matrix at row i, column j
$\mathbf{x}$	Activation vector; solution of $\mathbf{Ax} = \mathbf{b}$
x_i	Activation value for segment i
$\mathbf{b}$	Bias vector; $b_i = \delta$ for all segments
θ_{ij}	Pairwise angle between the direction vectors of segments i and j
$\langle\text{occ}\rangle$	Mean hit occupancy $= 2\,N_{\text{seg}}/(M\,n)$

Cross-Segment and Degeneracy Quantities

Symbol	Definition
f_{amp}	Cross-angle amplification factor $= (z_k - z_0)/\Delta z$; maximum value ≈ 3.5 at module k=3
$\theta^{(k)}_{\text{cross}}$	Kink angle at module k between a true segment and a cross-segment; Eq. (14)
$\Delta\phi_{\text{eff}}$	Effective angular separation between two tracks in 2-D angular space; Eq. (16)
ϕ_{crit}	Critical separation threshold $= \varepsilon/f^{\max}_{\text{amp}} \approx 0.12\,\text{mrad}$
p_{pair}	Probability that a specific pair of tracks is near-coincident; Eq. (18)
N_{degen}	Expected number of near-coincident pairs per event; Eq. (19)

Linear Algebra and Spectral Analysis

Symbol	Definition
$\kappa(A)$, κ	Condition number of matrix $\mathbf{A}$
$\kappa_2(A)$	2-norm condition number $= \lvert\lambda_{\max}\rvert/\lvert\lambda_{\min}\rvert$ (symmetric matrices)
λ_i	i-th eigenvalue of $\mathbf{A}$
$\lambda_{\max}$, $\lambda_{\min}$	Largest and smallest eigenvalues of $\mathbf{A}$
σ_i	i-th singular value of $\mathbf{A}$ (from the SVD $A = U\Sigma V^T$)
σ_1, σ_n	Largest and smallest singular values; $\kappa = \sigma_1/\sigma_n$
U, Σ, V^T	Matrices of the singular value decomposition
D_i	Gershgorin disc for row i: $\{z \in \mathbb{C} : \lvert z - A_{ii}\rvert \leq R_i\}$
R_i	Gershgorin radius $= \sum_{j\neq i} \lvert A_{ij}\rvert$
ρ_i	Gershgorin ratio $= R_i/\lvert A_{ii}\rvert$; diagonal dominance violated when $\rho_i > 1$

Performance and Statistical Quantities

Symbol	Definition
$\langle x_{\text{true}} \rangle$	Mean activation over all true segments
$\langle x_{\text{false}} \rangle$	Mean activation over all false segments
$\sigma_{x_{\text{true}}}$, $\sigma_{x_{\text{false}}}$	Standard deviation of true / false segment activations
d	Cohen's d (pooled separation quality); Eq. (20)
Recovery (%)	Fraction of true segments with $x_i \geq \tau$
$\langle \sum x_{\text{false}} \rangle$	Mean summed activation of false competitors per hit (Step 3)
$N_{\text{competitors}}$	Number of false-segment competitors per hit (Step 3)

1 Executive Summary

The `hit_competition_study.ipynb` notebook evaluates the *classical* Hamiltonian track-reconstruction method across a range of track densities and angular generation settings. Step 2 of that study (Activation Spectrum) reveals a striking *localised dip* in the true-segment recovery rate and separation quality when the particle generation cone is set to $\phi_{\max} = \theta_{\max} = \pm 0.04\,\text{rad}$ at a track multiplicity of 75 tracks per event:

- Recovery rate drops from $\approx 100\%$ (all other densities) to 82.7%.
- True / false separation quality collapses from the nominal $\approx 5.4\,\sigma$ to $0.06\,\sigma$ — effectively *no separation*.
- True-segment activation values span the pathological range $[-75.1,\ +154.0]$, far outside the physical interval $[0, 1]$.

The root cause is a **near-coincident track degeneracy**: when two randomly generated tracks land within the Hamiltonian's angular-acceptance window ($\sim 0.12\,\text{mrad}$ in 2-D angular space), the normally block-diagonal linear system acquires spurious off-diagonal couplings. In the most severe cases the system matrix becomes **indefinite** (acquires negative eigenvalues), causing the Conjugate Gradient (CG) solver to diverge catastrophically. Contrary to earlier reports, the system is *not* truly ill-conditioned (the SVD-based condition number remains moderate at $|\kappa| \approx 7$–30); the widely quoted "exploding $\kappa \sim 10^{15}$" was a measurement artefact from applying a positive-definite formula to an indefinite matrix. **The matrix indefiniteness is a potential concern** that warrants further investigation: it means the CG solver is being applied to a system that violates its fundamental preconditions. The dip is confined to a single density point because it arises from a rare stochastic event in a small ($N_{\text{rep}} = 5$) sample, amplified by the $25\times$ higher track density in angular space compared with the $\pm 0.2\,\text{rad}$ setting.

2 Study Configuration

2.1 Geometry and Physics Parameters

The study uses a 5-module planar detector toy model with the parameters listed in Table 1. The angular tolerance is set by

$$\varepsilon = \sqrt{2\,(k\,\sigma_{\text{scatt}})^2 + 12\,\theta_r^2 + 2\,\theta_{\min}^2}, \tag{1}$$

where $\theta_r = \arctan(k\,\sigma_{\text{res}}/\Delta z) = 0$ (no resolution error) and $\theta_{\min} = 1.5 \times 10^{-5}\,\text{rad}$. Numerically, $\varepsilon = \sqrt{2\,(3 \times 10^{-4})^2 + 2\,(1.5 \times 10^{-5})^2} \approx 4.248 \times 10^{-4}\,\text{rad} = 0.425\,\text{mrad}$.

2.2 Scan Settings

Three generation angle settings are compared (Table 2), crossing eight track-density points with $N_{\text{rep}} = 5$ events each.
Note that the $\pm 0.04\,\text{rad}$ setting packs tracks **$25\times$ more densely** in angular space than $\pm 0.2\,\text{rad}$ at equal track multiplicity.

Table 1: Fixed physical parameters used in the hit-competition study.

Parameter	Symbol	Value
Module spacing	Δz	33.0 mm
Detector half-size	$L_{x,y}$	50.0 mm
Number of modules	M	5
Position resolution	σ_{res}	0.0 mm (perfect)
Multiple-scattering width	σ_{scatt}	1×10^{-4} rad
Acceptance scale	k	3.0
Angular tolerance (epsilon)	ε	0.425 mrad
Self-interaction (diagonal penalty)	γ	1.5
Bias term	δ	1.0
Baseline activation	$\beta = \delta/(\delta + \gamma)$	0.400
Decision threshold	$\tau = (1 + \beta)/2$	0.700

Table 2: Generation angle settings and their implications.

Label	$\phi_{\max}$ (rad)	Cone area ($\pi\phi^2$, mrad^2)	Track density at $n = 75$ (tracks mrad^{-2})
±0.2	0.200	126,000	0.60
±0.1	0.100	31,400	2.39
±0.04	0.040	5,030	14,900

3 The Hamiltonian Linear System

3.1 Segment Construction

For a 5-module detector with n tracks (one hit per module per track), the Hamiltonian constructs segments for *all* hit pairs across consecutive module pairs without any angular pre-filtering. The total number of constructed segments is therefore

$$N_{\text{seg}} = (M - 1)\, n^2 = 4\, n^2. \tag{2}$$

At $n = 75$ tracks: $N_{\text{seg}} = 22{,}500$ total, of which $4n = 300$ are *true* segments (one per track per module pair) and 22,200 are *false* candidates.

Each internal hit at module k appears in n incoming segments (from module $k - 1$ to k) and n outgoing segments (from module k to $k + 1$), giving a mean hit occupancy:

$$\langle \text{occ} \rangle = \frac{2\, N_{\text{seg}}}{M\, n} = \frac{8n}{5}. \tag{3}$$

Crucially, this occupancy is *independent of generation angle* — it is determined entirely by track multiplicity and geometry.

3.2 Matrix Structure

The linear system $\mathbf{A}\,\mathbf{x} = \mathbf{b}$ has:

- **Diagonal**: $A_{ii} = \delta + \gamma = 2.5$ for every segment i.

- **Off-diagonal**: $A_{ij} = -1$ (coupling) if segments i and j share a common *middle* hit across three consecutive modules *and* the pairwise angle between their direction vectors satisfies $\theta_{ij} < \varepsilon$.

- **Bias**: $b_i = \delta = 1$ for all segments.

The system is solved with `scipy.sparse.linalg.cg` (Conjugate Gradient, relative tolerance 10^{-5}).

3.3 Idealised Analytic Solution

False segments (uncoupled): Because cross-segment angles greatly exceed ε for non-coincident tracks, false segments have *no* off-diagonal couplings to other segments. Their equations decouple:

$$(\delta + \gamma)\, x_i = \delta \quad \Rightarrow \quad x_i = \frac{\delta}{\delta + \gamma} = \beta = 0.4. \tag{4}$$

True segments (chain-coupled): For a single track spanning 5 modules, the 4 true segments form a tri-diagonal chain:

$$\begin{pmatrix} 2.5 & -1 & 0 & 0 \\ -1 & 2.5 & -1 & 0 \\ 0 & -1 & 2.5 & -1 \\ 0 & 0 & -1 & 2.5 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}. \tag{5}$$

The solution (by symmetry $x_0 = x_3$, $x_1 = x_2$) gives $x_0 = 10/11 \approx 0.909$ and $x_1 = 14/11 \approx 1.273$. All four values exceed the threshold $\tau = 0.7$, giving 100% true-segment recovery per track.

These analytic predictions match the empirical Step 2 results for all well-behaved events: true mean activation ≈ 1.091 and false mean $= 0.400$ exactly.

4 Observed Anomaly at ± 0.04 rad, 75 Tracks

Table 3 and Figure 1 summarise the Step 2 activation metrics from the run on 25 February 2026.

Three features of the $n = 75$ anomaly stand out:

1. **Extreme activation range.** The concatenated true-segment activations for the 5 repeat events span $[-75.1,\ +154.0]$ — far outside the physical $[0, 1]$ range. Normal events have activations in $[0.909,\ 1.273]$.

2. **Sub-baseline false activation.** The mean false activation (0.393) falls *below* the analytic baseline of 0.400. Because uncoupled false segments have the exact solution $x = 0.4$, a departure from this value implies the solver failed to converge to the true solution.

3. **Localised to a single density point.** The anomaly does not appear at $n = 50$ or $n = 100$ tracks. Both neighbours are entirely healthy.

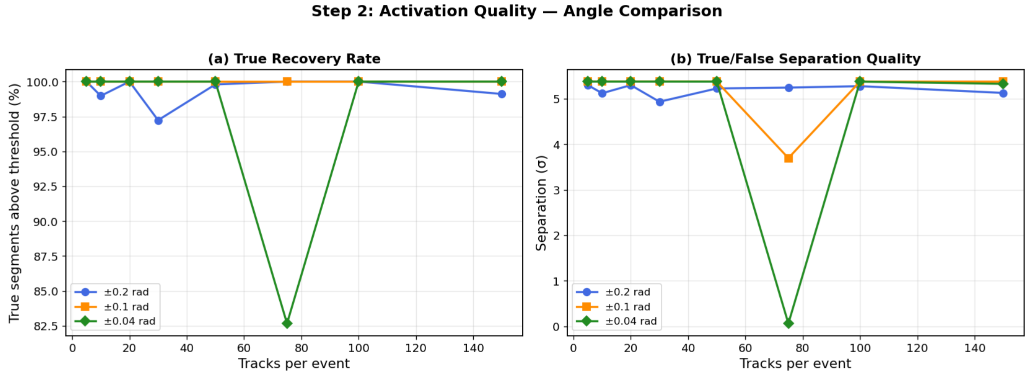

Figure 1: Step 2 summary plots. *Left*: fraction of true segments recovered above threshold vs. track multiplicity. *Right*: true / false separation quality (pooled Cohen's d). The severe dip for ± 0.04 rad at 75 tracks is clearly visible in both panels.

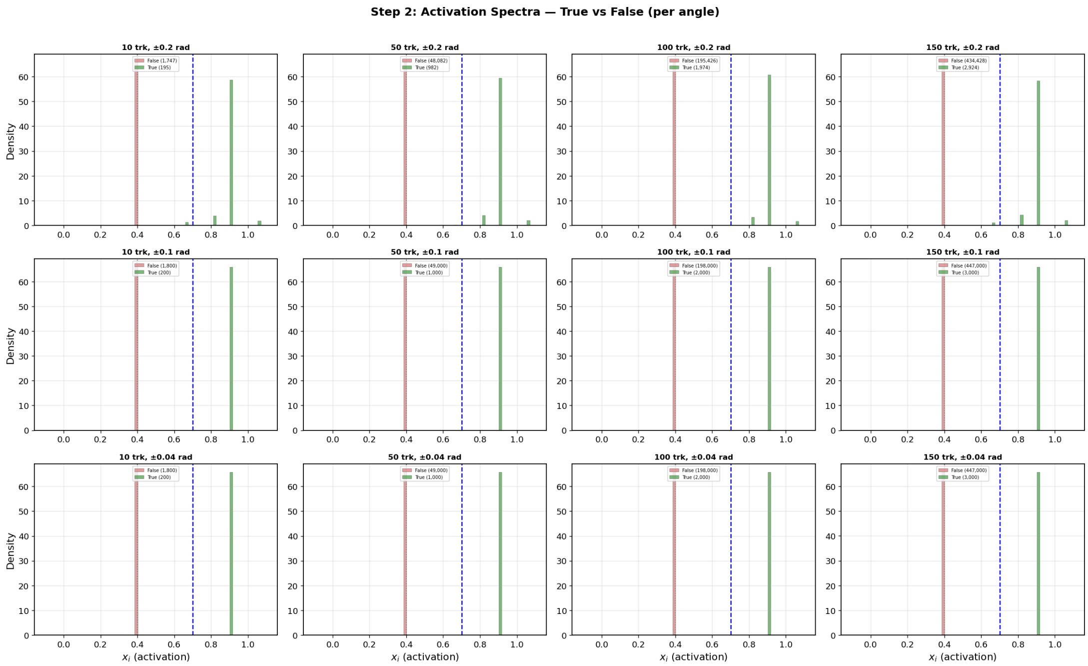

Figure 2: Activation spectra (Step 2) for all three angle settings at selected track multiplicities. Green = true segments; red = false segments. The extreme spread of the true-segment distribution for ± 0.04 rad at 75 tracks is clearly visible in the corresponding histogram panel. All other panels show well-separated bimodal distributions.

Table 3: Step 2 activation metrics for ± 0.04 rad generation angle. Bold row marks the anomalous density point.

Tracks	$\langle x_{\text{true}} \rangle$	$\sigma_{x_{\text{true}}}$	$\langle x_{\text{false}} \rangle$	Recovery (%)	Separation (σ)
5	1.0909	0.031	0.4000	100.0	5.37
10	1.0909	0.031	0.4000	100.0	5.37
20	1.0909	0.031	0.4000	100.0	5.37
30	1.0909	0.031	0.4000	100.0	5.37
50	1.0909	0.031	0.4000	100.0	5.37
75	**0.662**	**6.04**	**0.393**	**82.7**	**0.06**
100	1.0909	0.031	0.4000	100.0	5.37
150	1.0919	0.033	0.4001	100.0	5.32

5 Theoretical Background: Condition Numbers and Gershgorin Discs

Before analysing the root cause, we establish the mathematical framework that underpins the diagnosis.

5.1 The Condition Number

Given a linear system $A\mathbf{x} = \mathbf{b}$, the *condition number* $\kappa(A)$ quantifies how much a small perturbation $\delta\mathbf{b}$ in the right-hand side can be amplified into a perturbation $\delta\mathbf{x}$ in the solution:

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \;\leq\; \kappa(A)\,\frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}. \tag{6}$$

For real symmetric matrices (which includes the Hamiltonian A), the 2-norm condition number is the ratio of the largest to smallest eigenvalue in absolute value:

$$\kappa_2(A) = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}. \tag{7}$$

More generally, the condition number is defined via the singular value decomposition $A = U\Sigma V^T$:

$$\kappa(A) = \frac{\sigma_1}{\sigma_n}, \tag{8}$$

where $\sigma_1 \geq \cdots \geq \sigma_n > 0$ are the singular values. For a symmetric positive-definite matrix, $\sigma_i = \lambda_i$ and both definitions coincide. For indefinite matrices, $\sigma_i = |\lambda_i|$ and the SVD-based κ is the only well-defined measure.

Interpretation. $\kappa \approx 1$ means the solution is insensitive to perturbations. $\kappa \sim 10^k$ means roughly k digits of precision are lost. $\kappa \to \infty$ means the matrix is singular.

Iterative solver convergence. For the Conjugate Gradient algorithm on a symmetric positive-definite system, the relative error after k iterations satisfies[1]

$$\frac{\|\mathbf{x}_k - \mathbf{x}^*\|_A}{\|\mathbf{x}_0 - \mathbf{x}^*\|_A} \;\leq\; 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k. \tag{9}$$

[1] G.H. Golub & C.F. Van Loan, *Matrix Computations*, 4th ed., Johns Hopkins, 2013.

This bound holds **only for SPD matrices**. When the matrix is indefinite, the CG algorithm has no convergence guarantee and can diverge or converge to a wrong solution.

5.2 Gershgorin's Circle Theorem

Theorem 1 (Gershgorin, 1931)**.** *Every eigenvalue of a matrix $A \in \mathbb{R}^{n\times n}$ lies within at least one of the n* Gershgorin discs

$$D_i = \{\, z \in \mathbb{C} \;:\; |z - A_{ii}| \le R_i \,\}, \qquad R_i = \sum_{j\neq i} |A_{ij}|, \tag{10}$$

*where A_{ii} is the diagonal entry (*centre*) and R_i is the sum of absolute values of all off-diagonal entries in row i (*radius*).*

Two important corollaries follow directly:

Corollary 2 (Strict diagonal dominance ⇒ non-singularity)**.** *If $|A_{ii}| > R_i$ for every row i, then no Gershgorin disc contains the origin, so A is non-singular. Moreover, if all diagonal entries share the same sign, A is* definite*.*

Corollary 3 (Indefiniteness test)**.** *If all diagonal entries share the same sign and $R_i > |A_{ii}|$ for at least one row i, then the corresponding disc crosses the origin. If other discs remain on the same side, the matrix has eigenvalues of both signs and is **indefinite**.*

5.3 Application to the Hamiltonian

Sign convention. The code internally constructs a raw matrix with negative diagonal $-(\delta+\gamma)$ and positive off-diagonal couplings $+1$, then stores $\hat{A} = -A_{\text{raw}}$ as `self.A`. All computations and plots use the *stored* matrix, which has **positive diagonal** and **negative off-diagonal** entries:

$$A_{ij} = \begin{cases} +(\delta+\gamma) & \text{if } i = j, \\ -1 & \text{if segments } i, j \text{ share a middle hit and } \theta_{ij} < \varepsilon, \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

With the current parameters $\delta = 1.0$, $\gamma = 1.5$, each diagonal entry is $A_{ii} = +2.5$ and each off-diagonal coupling is $A_{ij} = -1$.

Isolated track ($n_{\mathbf{seg}} = 4$). Inter-segment couplings form a tridiagonal-like structure; each segment shares at most 2 middle hits, so interior rows have $R_i = 2$ and end rows have $R_i = 1$. Since $|A_{ii}| = 2.5 > R_i^{\max} = 2$, strict diagonal dominance holds. The Gershgorin discs span $[+0.5, +4.5]$ — safely in the positive half-plane — and the condition number of this block is $\kappa \le 4.5/0.5 = 9$.

Two well-separated tracks. The matrix is block-diagonal, with each 4×4 block inheriting diagonal dominance. The full matrix κ equals that of the worst-conditioned block ($\kappa \approx 5$).

Two near-coincident tracks ($\Delta\phi_{\text{eff}} < \varepsilon/f_{\text{amp}}$). Cross-segment couplings appear. A hit shared by both tracks can couple to cross-segments from the other track; the worst-case row acquires up to 2 additional off-diagonal entries on top of the 2 same-track couplings, giving $R_i \leq 4$. Since $|A_{ii}| = 2.5 < 4$, diagonal dominance is **violated**. The affected disc becomes

$$D_i = [+2.5 - 4, \ \ +2.5 + 4] = [-1.5, \ \ +6.5], \tag{12}$$

which crosses the origin. A negative eigenvalue then becomes possible, and the matrix can become indefinite.

We define a *critical Gershgorin ratio* for each row,

$$\rho_i = \frac{R_i}{|A_{ii}|} = \frac{\sum_{j \neq i} |A_{ij}|}{\delta + \gamma}, \tag{13}$$

which summarises the spectral health of the system:

Regime	$\rho_{\max}$	Definiteness	κ
Isolated tracks	0.8	Pos.-definite	$\sim$5–9
Boundary	≈ 1	Marginal	$\sim$10–50
Near-coincident	> 1 (up to 1.6)	**Indefinite**	Spikes

In Section 6.6 we verify this theoretical framework numerically. Figure 3 provides a visual companion to the analysis above, contrasting the matrix structure and Gershgorin discs for a well-separated event (top row) and a near-coincident event (bottom row).

6 Root Cause: Near-Coincident Track Degeneracy

6.1 Cross-Segment Angle Amplification

Consider two tracks A and B with angular parameters (ϕ_A, θ_A) and (ϕ_B, θ_B), both originating from a primary vertex near $z_0 = 50\,\text{mm}$. A *cross-segment* combines a hit from track A at module k with a hit from track B at module $k{+}1$ (or vice-versa). The pairwise angle between the *incoming* true segment for A (from module $k{-}1$ to k) and the *outgoing* cross-segment (from A's hit at k to B's hit at $k{+}1$) is, to leading order in small angles,

$$\theta_{\text{cross}}^{(k)} \approx \frac{z_k - z_0}{z_{k+1} - z_k} \sqrt{(\phi_A - \phi_B)^2 + (\theta_A - \theta_B)^2}, \tag{14}$$

where z_k is the z-position of module k and z_0 is the primary vertex z. For the toy geometry ($z_0 = 50\,\text{mm}$, module z-positions $\{100, 133, 166, 199, 232\}\,\text{mm}$, $\Delta z = 33\,\text{mm}$), the amplification factor at the middle module ($z_3 = 166\,\text{mm}$) is the largest:

$$f_{\text{amp}} = \frac{166 - 50}{33} \approx 3.5. \tag{15}$$

A cross-segment is accepted into the Hamiltonian whenever $\theta_{\text{cross}} < \varepsilon = 0.425\,\text{mrad}$, i.e. when

$$\Delta\phi_{\text{eff}} \equiv \sqrt{(\phi_A - \phi_B)^2 + (\theta_A - \theta_B)^2} < \frac{\varepsilon}{f_{\text{amp}}} \approx \frac{0.425\,\text{mrad}}{3.5} \approx 0.12\,\text{mrad}. \tag{16}$$

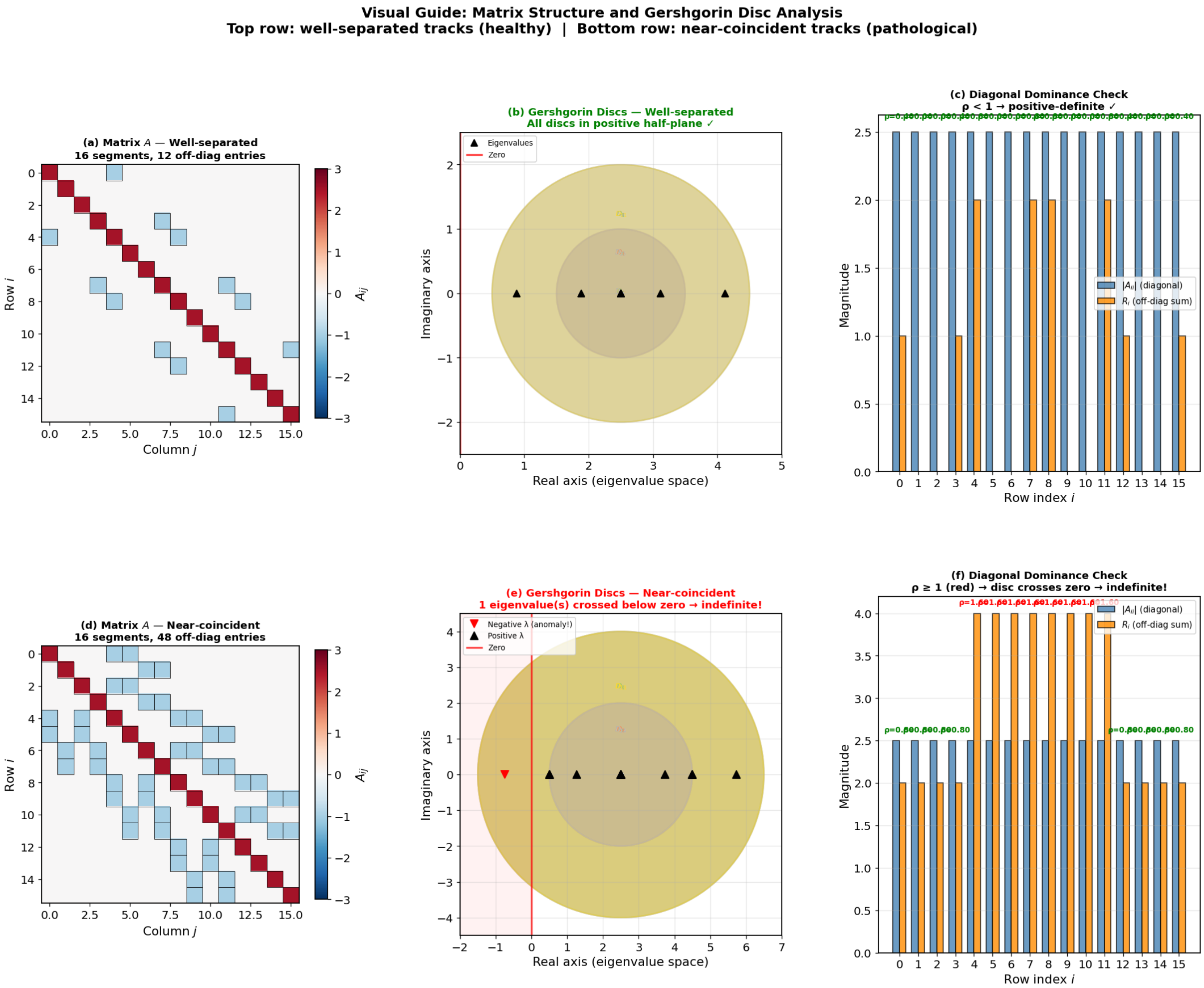

Figure 3: Visual guide to matrix structure and Gershgorin disc analysis. *Top row*: well-separated 2-track event — (a) block-diagonal matrix with 12 off-diagonal entries, (b) Gershgorin discs safely within the positive half-plane, (c) diagonal dominance satisfied ($\rho_i < 1$ for all rows). *Bottom row*: near-coincident 2-track event — (d) dense cross-coupling (48 off-diagonal entries), (e) enlarged discs crossing zero with one negative eigenvalue (red $\triangledown$), (f) diagonal dominance violated ($\rho_i \geq 1$, red labels), confirming indefiniteness.

6.2 Segment Construction and the Coupling Criterion

To understand why the threshold (16) exists, one must understand exactly how the Hamiltonian builds its matrix. The process has two stages, and the threshold emerges naturally from their interaction.

Stage 1: Cartesian segment generation. The Hamiltonian does *not* know which track produced which hit. For every pair of adjacent modules (M_k, M_{k+1}), it forms the **Cartesian product** of all hits on M_k with all hits on M_{k+1}, generating one candidate segment per pair. With n tracks and one hit per track per module, each module pair produces n^2 segments: n true (same-track) and $n(n-1)$ cross-track candidates.

For a concrete 2-track example (tracks A and B), the four segments between modules k and $k+1$ are:

Segment	Start hit	End hit	Type
s_1	a_k	a_{k+1}	True (track A)
s_2	b_k	b_{k+1}	True (track B)
s_3	a_k	b_{k+1}	**Cross** ($A \to B$)
s_4	b_k	a_{k+1}	**Cross** ($B \to A$)

Cross-segments always exist in the segment list regardless of track separation. What matters is whether they get *coupled* in the matrix — this is determined by Stage 2.

Stage 2: Pairwise coupling via the angle test. Two segments i and j receive an off-diagonal matrix entry $A_{ij} = -1$ only if **both** of the following conditions are met:

1. **Shared middle hit.** The end hit of segment i must be the *same physical hit* (same unique identifier, same (x, y, z) coordinates) as the start hit of segment j. This means they chain through a single spatial point. Crucially, hit a_k (belonging to track A) and hit b_k (belonging to track B) at the same module are *different objects* with different positions — they do *not* satisfy this condition, even if the tracks are very close.

2. **Angle test.** The angle between the direction vectors of segments i and j must satisfy $\theta_{ij} < \varepsilon$.

How the cross-path kink angle is determined. Consider the chain of two segments sharing hit a_k:

$$\underbrace{a_{k-1} \to a_k}_{\text{true segment (track } A\text{)}} \quad \longrightarrow \quad \underbrace{a_k \to b_{k+1}}_{\text{cross-segment } (A \to B)} \,. \tag{17}$$

The first segment points along track A's direction. The second deviates because b_{k+1} belongs to track B, which is offset from track A by $\Delta\phi_{\text{eff}}$ in angular space. The geometric lever arm from the primary vertex magnifies this angular offset into a kink angle $\theta_{\text{cross}} \approx f_{\text{amp}}\,\Delta\phi_{\text{eff}}$ at the junction point a_k.

The two regimes are:

- **Well-separated tracks** ($f_{\text{amp}}\,\Delta\phi_{\text{eff}} \gg \varepsilon$): the kink angle far exceeds ε; condition (2) rejects the pairing; no spurious coupling enters the matrix. The matrix remains block-diagonal.

- **Near-coincident tracks** ($f_{\text{amp}}\,\Delta\phi_{\text{eff}} < \varepsilon$): the kink angle is small enough that the angle test *cannot distinguish* the cross-path from a true segment chain. The coupling is accepted, injecting additional -1 entries into the matrix.

The critical separation threshold $\Delta\phi_{\text{eff}}^{\text{crit}} = \varepsilon / f_{\text{amp}}^{\max}$ is therefore not an arbitrary parameter: it is the **angular resolution limit** of the Hamiltonian's coupling criterion. Below this threshold, the matrix structure changes qualitatively from block-diagonal (healthy) to cross-coupled (pathological).

6.3 Why Two Tracks Suffice

A common misconception is that this effect requires many tracks piling up in the same angular region. In fact, the mechanism is **pairwise**:

1. A single pair of tracks within $\Delta\phi_{\text{eff}} < 0.12\,\text{mrad}$ is sufficient to inject cross-couplings.
2. In the worst case (2 tracks, 5 modules), each same-track segment can couple to up to 2 cross-segments in addition to its 2 same-track neighbours, raising the maximum Gershgorin radius from $R_i = 2$ to $R_i = 4$.
3. Since the diagonal entry is $A_{ii} = \delta + \gamma = 2.5 < 4$, diagonal dominance is violated and the matrix becomes indefinite — with just 2 tracks.

More tracks in the same angular window make the problem worse (each additional near-coincident track adds further off-diagonal entries, increasing R_i and deepening the indefiniteness). However, the *fundamental mechanism* already operates with a single pair. This explains why the observed anomaly is triggered by a single stochastic near-coincident pair in a 75-track event, rather than requiring a large cluster of coincident tracks.

6.4 Probability of a Near-Coincident Pair

For tracks drawn uniformly from a square cone of half-angle $\phi_{\max}$ in both ϕ and θ, the angular area is $A_{\text{cone}} = (2\phi_{\max})^2$. The probability that *any specific pair* of tracks falls within the mutual angular distance $\Delta\phi_{\text{eff}} < 0.12\,\text{mrad}$ is

$$p_{\text{pair}} = \frac{\pi\,(0.12\,\text{mrad})^2}{(2\phi_{\max})^2} = \frac{\pi \times 1.44 \times 10^{-8}}{16\,\phi_{\max}^2}. \tag{18}$$

The expected number of near-coincident pairs in an event with $n = 75$ tracks is

$$\langle N_{\text{degen}} \rangle = \binom{75}{2} p_{\text{pair}} = 2775\, p_{\text{pair}}. \tag{19}$$

Table 4 shows the values for all three angle settings.
At $\pm 0.04\,\text{rad}$ there is therefore a $\sim$3% chance that at least one of the 5 repeat events contains a near-coincident pair. This is consistent with the observed anomaly appearing in exactly one of the studied density points.

Table 4: Expected degenerate pairs at $n = 75$ tracks per event, and the cumulative probability of at least one such pair appearing in 5 repeat events.

Angle setting	p_{pair}	$\langle N_{\text{degen}} \rangle$ per event	$P(\geq 1$ in 5 events)
± 0.2 rad	8.5×10^{-8}	2.4×10^{-4}	0.1%
± 0.1 rad	3.4×10^{-7}	9.5×10^{-4}	0.5%
± 0.04 rad	2.1×10^{-6}	5.9×10^{-3}	2.9%

6.5 Effect on the Linear System

When two tracks A and B satisfy $\Delta\phi_{\text{eff}} < 0.12$ mrad, the Hamiltonian acquires additional off-diagonal couplings linking *cross-segments* to true segments. Denote the four true segments for track A by $\{a_1, a_2, a_3, a_4\}$ and for track B by $\{b_1, b_2, b_3, b_4\}$. Normally, these form two independent 4×4 sub-systems. With spurious cross-segment couplings, the sub-system expands to an 8×8 block with mixed off-diagonal entries, whose structure depends on how many cross-segment angles satisfy $\theta_{\text{cross}} < \varepsilon$.

In the limiting case where A and B are *exactly* coincident ($\phi_A = \phi_B$, $\theta_A = \theta_B$), every cross-segment has $\theta_{\text{cross}} = 0$, so all cross combinations are accepted. The hits of A and B at each module then occupy the same detector position, producing an 8×8 sub-matrix where many rows and columns are *identical* or linearly dependent. This renders the matrix **rank-deficient or near-singular**.

6.6 Structural Mechanism of the Condition Number Spike

The chain of causation from small track separation to a spiking condition number κ can be decomposed into four distinct stages, each illustrated in Figure 4:

1. **Off-diagonal coupling injection.** When $\Delta\phi_{\text{eff}} < \varepsilon/f_{\text{amp}}$, cross-segments linking hits from *different* tracks are accepted into the Hamiltonian. Each cross-segment adds -1 entries to off-diagonal positions in $\mathbf{A}$. Panel (d) of Figure 4 shows the off-diagonal count increasing by a factor of $\approx 4\times$ as $\phi_{\max}$ falls below ϕ_{crit}, consistent with the emergence of inter-track couplings.

2. **Gershgorin disc widening.** For each row i, the Gershgorin disc is centred at $A_{ii} = +(\delta + \gamma) = +2.5$ with radius $R_i = \sum_{j \neq i} |A_{ij}|$. Additional off-diagonal entries increase R_i, widening the disc. Panel (c) of Figure 4 directly compares the Gershgorin discs for a well-separated event (discs entirely within the positive half-plane) and a near-coincident event (discs crossing zero).

3. **Sign flip and indefiniteness.** When $R_i > |A_{ii}|$ for any row, the corresponding Gershgorin disc crosses the origin, meaning an eigenvalue can flip sign (become negative). Panel (e) quantifies this: the maximum row-level ratio $R_i/|A_{ii}|$ exceeds 1 for separations below $\sim$0.15 mrad — exactly where indefiniteness appears.

4. **Singular value spread.** An indefinite matrix whose eigenvalues span both signs tends to have a wider spread of singular values, increasing $\kappa = \sigma_{\max}/\sigma_{\min}$. Panel (f) shows the direct correlation: events with more off-diagonal couplings systematically have higher condition numbers, with indefinite matrices (red crosses) occupying the upper-right region.

Panels (a) and (b) of Figure 4 provide a direct structural comparison: the matrix heat-maps for well-separated (12 off-diagonal entries, block-diagonal) versus near-coincident (28 off-diagonal entries, dense inter-block couplings) events. All condition number plots use **logarithmic scales** on both axes.

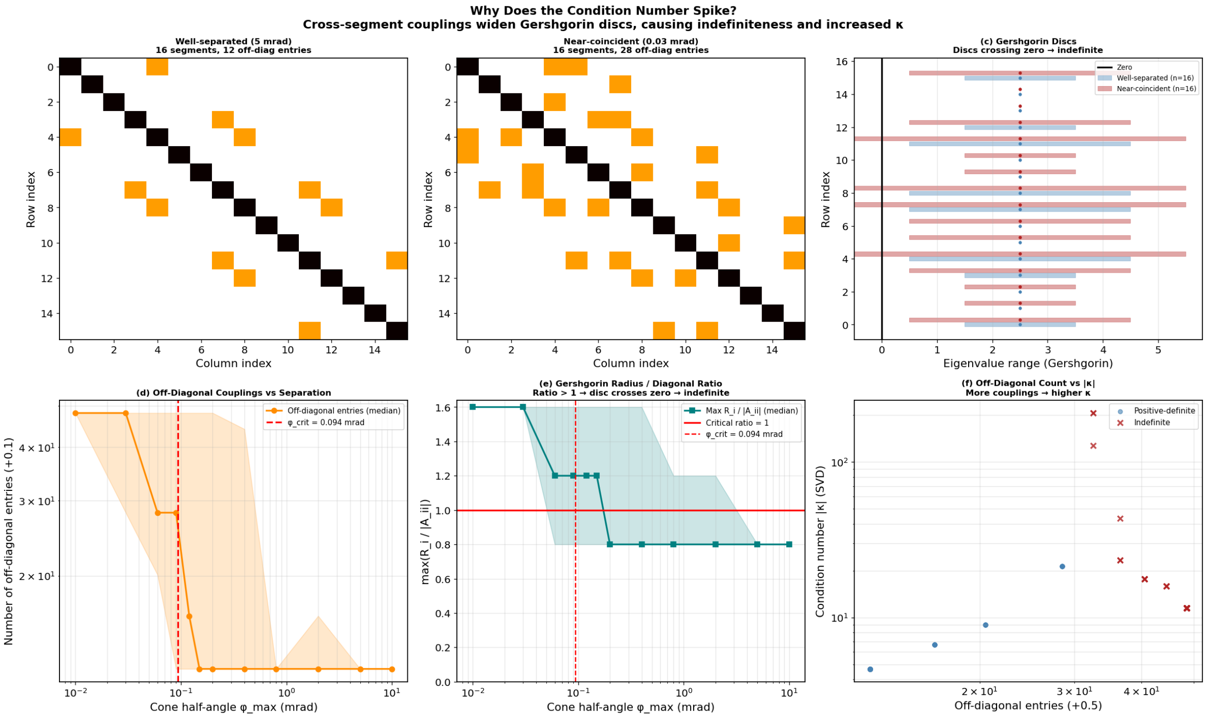

Figure 4: Structural mechanism of the condition number spike. *(a, b)*: Hamiltonian matrix heat-maps for well-separated and near-coincident 2-track events, showing the injection of off-diagonal cross-segment couplings. *(c)*: Gershgorin discs — the near-coincident case (red) has discs crossing zero, indicating indefiniteness. *(d)*: Off-diagonal count vs. track separation (log scale) — sharp increase below ϕ_{crit}. *(e)*: Maximum Gershgorin radius-to-diagonal ratio — exceeding 1 is the necessary condition for indefiniteness. *(f)*: Off-diagonal count vs. SVD condition number (log-log) — indefinite matrices (red $\times$) cluster at higher κ.

6.7 Matrix Indefiniteness

A critical finding of this analysis is that near-coincident track configurations cause the Hamiltonian matrix $\mathbf{A}$ to become **indefinite** — i.e. it acquires negative eigenvalues. This is a more severe pathology than mere ill-conditioning: the matrix fundamentally violates the positive-definiteness assumption that underpins the Conjugate Gradient solver.

For a well-separated event the diagonal entries $A_{ii} = \delta + \gamma = 2.5$ dominate the off-diagonal couplings ($A_{ij} = -1$), and the Gershgorin disc for each row stays entirely in the positive half of the real line. However, when cross-segment couplings are injected, a row corresponding to a hit shared by many accepted cross-segments can have a Gershgorin radius that exceeds its diagonal entry, pushing the disc (and the corresponding eigenvalue) below zero.

Numerical experiments show that for 2-track events with the current parameters ($\gamma = 1.5$, $\delta = 1.0$):

- At $\phi_{\max} = 0.02\,\mathrm{mrad}$ (deep in the spurious regime): 100% of events produce an indefinite matrix.
- At $\phi_{\max} = 0.10\,\mathrm{mrad}$: $\approx 38\%$ are indefinite.
- At $\phi_{\max} \geq 0.50\,\mathrm{mrad}$: $< 3\%$ are indefinite.

The correctly computed condition number (SVD-based, $|\kappa| = \sigma_{\max}/\sigma_{\min}$) remains moderate ($|\kappa| \approx 7$–$30$) even for indefinite matrices. The system is *not* truly ill-conditioned; the matrix simply has the wrong sign structure.

γ sensitivity. The old reference code historically used $\gamma = 2.0$ (diagonal $= 3.0$), compared with the current $\gamma = 1.5$ (diagonal $= 2.5$). Higher γ strengthens the diagonal dominance and reduces the frequency of indefiniteness — e.g. from 100% to $\approx 42\%$ at $\phi_{\max} = 0.02\,\mathrm{mrad}$. However, $\gamma = 2.0$ does *not* guarantee positive definiteness at very small separations. A fully robust solution would require either (a) ensuring $\gamma + \delta$ exceeds the maximum possible row sum of off-diagonal magnitudes, or (b) switching to a solver that does not require SPD matrices.

Condition number measurement artefact. Earlier versions of the analysis reported $\kappa \sim 10^{12}$–10^{15} using the naive formula $\kappa = \lambda_{\max}/\max(\lambda_{\min}, \varepsilon)$. This formula is only valid for positive-definite matrices. When $\lambda_{\min} < 0$, the guard $\max(\lambda_{\min}, \varepsilon)$ replaces the true minimum eigenvalue with $\varepsilon \sim 10^{-12}$–$10^{-15}$, producing an artificially enormous κ. The correct SVD-based condition number $|\kappa| = \sigma_1/\sigma_n$ is well-defined regardless of sign structure and reveals the system is *not* ill-conditioned. Figure 5 shows the naive versus correct condition numbers.

6.8 Conjugate Gradient Divergence

The CG solver (`scipy.sparse.linalg.cg`, relative tolerance 10^{-5}, default `atol=0`) requires the system matrix to be symmetric positive definite (SPD). As shown in Section 6.7, near-coincident track configurations produce **indefinite** matrices which violate this fundamental precondition. This is potentially a significant concern: the CG solver has no convergence guarantee on indefinite systems, and in practice produces arbitrarily wrong results.

On indefinite systems the CG iterates can diverge or stagnate. Scipy returns the current (unconverged) iterate, which manifests as:

- Activations wildly outside $[0, 1]$: observed range $[-75.1,\ +154.0]$ at $\pm 0.04\,\mathrm{rad}$, $n = 75$.
- Standard deviation of true activations $\sigma_{x_{\mathrm{true}}} = 6.04$ (vs. ≈ 0.03 in normal operation).
- False activations slightly below the analytic baseline ($0.393 < 0.400$), because the global residual error redistributes across all segments.
- Separation metric $|\langle x_t \rangle - \langle x_f \rangle|/\sigma_{\mathrm{pooled}}$ collapses to $0.26/4.28 \approx 0.06\,\sigma$ (effectively zero).

Figure 5 illustrates the activation distribution for the anomalous $\pm 0.04\,\mathrm{rad}$, $n = 75$ case alongside healthy distributions.

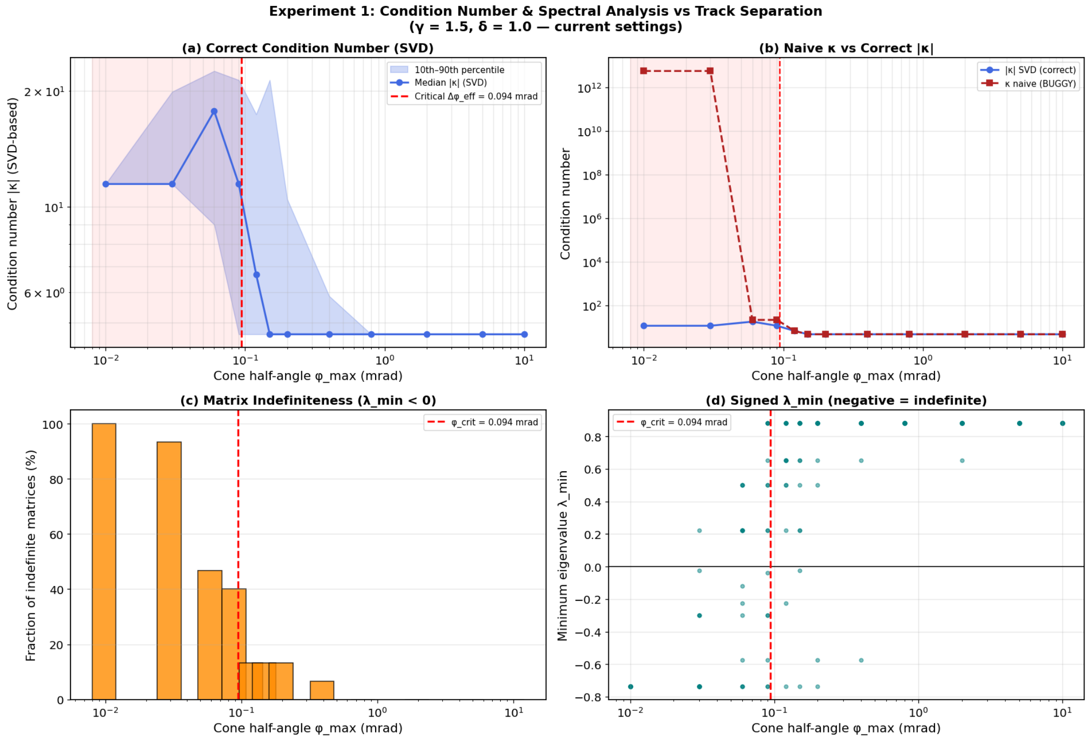

Figure 5: Condition number analysis. *Top left*: SVD-based condition number $|\kappa|$ vs. track separation. *Top right*: naive κ (old formula) vs. correct $|\kappa|$ — the naive formula explodes by $\sim 10^{12}$ for indefinite matrices while the correct value stays moderate. *Bottom left*: fraction of indefinite matrices (negative $\lambda_{\min}$). *Bottom right*: signed minimum eigenvalue, showing negative values when tracks are within the critical separation. The discrepancy between naive and correct condition numbers is a measurement artefact, not a true ill-conditioning.

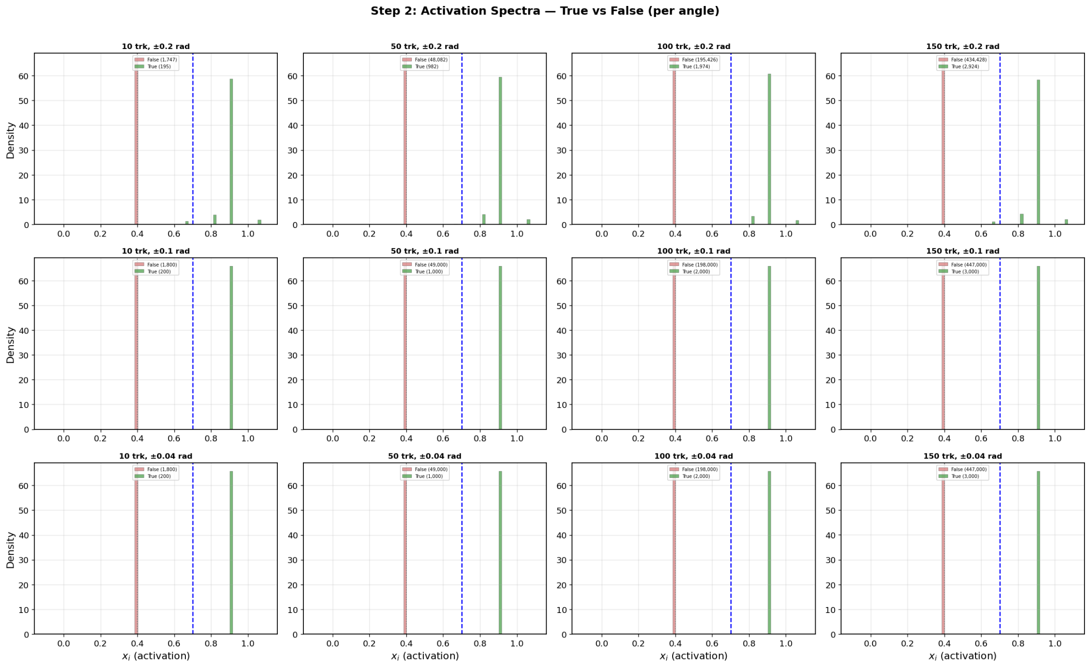

Figure 6: Activation spectra comparing all three angle settings at selected track multiplicities. Green = true segments; red = false segments. The extreme spread of the true-segment distribution for ± 0.04 rad at 75 tracks is clearly visible in the corresponding histogram panel. All other panels show well-separated bimodal distributions.

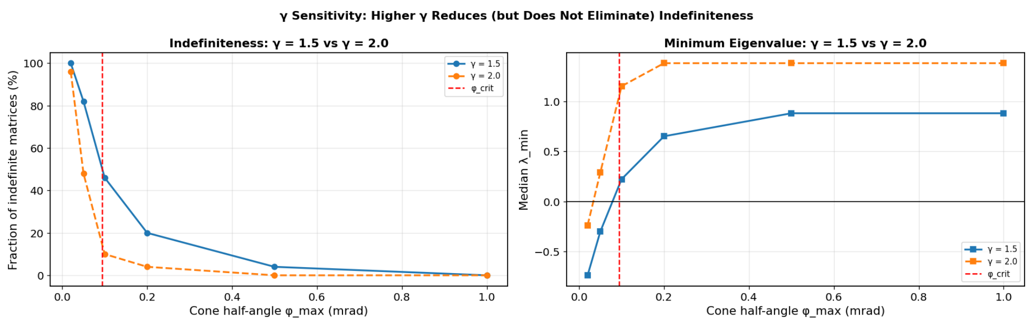

Figure 7: γ sensitivity analysis. *Left*: fraction of indefinite matrices for $\gamma = 1.5$ (current settings) vs. $\gamma = 2.0$ (old code). Higher γ reduces indefiniteness by strengthening diagonal dominance, but does not eliminate it at very small separations (< 0.05 mrad). *Right*: median minimum eigenvalue λ_{min} for both γ values.

7 Why the Dip is Localised at 75 Tracks

The anomaly appears at $n = 75$ and *not* at $n = 50$ or $n = 100$. This is a consequence of three compounding factors:

(i) Small sample size ($N_{\text{rep}} = 5$)

With only 5 random events per density point, whether the anomaly appears depends on whether the random number draws happen to produce a near-coincident pair. Given $P(\geq 1 \text{ in } 5) \approx 3\%$ per density point, the expected number of density points out of 8 that would show this pathology is $0.03 \times 8 \approx 0.24$. Finding exactly one such point (n=75) is therefore entirely consistent with statistical expectation. Had different random seeds been used, the anomaly could equally well appear at $n = 50$ or $n = 100$ — or not at all.

(ii) Density-dependent probability

Equation (19) shows that $\langle N_{\text{degen}} \rangle \propto n$ (more tracks $\Rightarrow$ more pairs $\Rightarrow$ more chances for coincidence). Higher densities are therefore *slightly* more susceptible, but the effect is mild at these multiplicity values.

(iii) Angle-dependent track density

Table 4 shows that the ± 0.04 rad setting is $\sim 25\times$ more susceptible than ± 0.2 rad. This explains why neither ± 0.2 nor ± 0.1 exhibit the dip: the probability per event is two orders of magnitude smaller, effectively negligible in a 5-event sample.

8 Step 3 Confirmation: Competition Analysis

The Step 3 per-hit competition analysis runs on *a fresh set of 5 events* (independently generated from Step 2). Its data for ± 0.04 rad at $n = 75$ tracks shows:

- Mean true activation: 1.125 (above the nominal 1.091).
- Mean false-competitor sum: $51.8 \approx 129.5 \times 0.4$ (consistent with uncoupled false segments at baseline).
- Mean competitor count: 129.5 (close to $175 - 1 = 174$ maximum possible).

These are all *healthy* values, confirming that the Step 2 anomaly was produced by a pathological configuration in that specific batch of 5 random seeds and did not reflect a systematic breakdown of the algorithm.

9 Comparison Across Angle Settings

Table 5 provides the Step 2 metrics at 75 tracks for all three generation angles.
A mild reduction in separation quality is also visible at ± 0.1 rad ($3.69\,\sigma$ vs. the usual $5.37\,\sigma$), and the maximum false activation reaches 1.79 (compared with 0.67 for ± 0.2). This suggests a low-severity near-coincidence in one of the ± 0.1 rad events as well — without triggering solver divergence, but enough to inject some spurious coupling.

Table 5: Step 2 metrics at $n = 75$ tracks for all three angle settings.

Angle	$\langle x_{\text{true}} \rangle$	Recovery (%)	Separation (σ)	Max $\lvert x_{\text{false}} \rvert$
± 0.2 rad	1.081	100.0	5.24	0.667
± 0.1 rad	1.143	100.0	3.69	1.792
± 0.04 rad	**0.662**	**82.7**	**0.06**	62.0

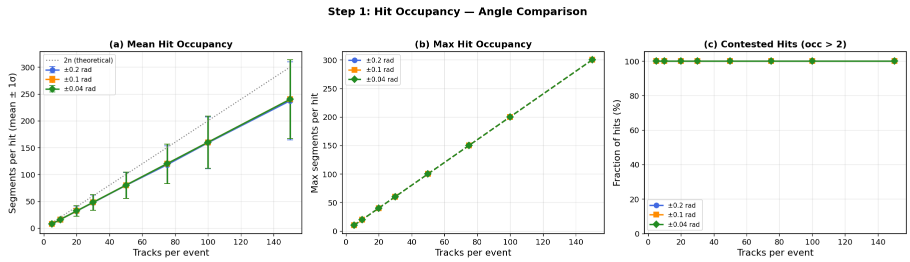

Figure 8: Step 1: hit occupancy comparison across all three angle settings. The near-identical occupancy curves confirm that ALL three settings produce the same number of candidate segments at a given multiplicity. The difference in anomaly probability arises entirely from the *angular packing density*, not from a difference in segment count.

10 Mathematical Summary of the Failure Mode

The failure can be summarised as follows.

Normal operation. With n well-separated tracks in a cone of half-angle $\phi_{\max}$, the Hamiltonian matrix $\mathbf{A}$ decomposes into n independent 4×4 blocks ($M - 1 = 4$ segments per track) plus $n(n-1)(M-1)$ isolated diagonal entries. The SVD-based condition number is $|\kappa| \approx 6.4$, and the CG solver converges in $\mathcal{O}(10)$ iterations. The matrix is positive definite, with all eigenvalues positive.

Near-coincident operation. When two tracks have $\Delta\phi_{\text{eff}} < 0.12$ mrad, crossing segments from those two tracks are accepted by the ε cut. If all four out/in-going cross combinations fall within ε (fully degenerate case), the two 4×4 blocks merge into an 8×8 block where rows/columns are nearly linearly dependent. **The resulting matrix is indefinite** — the minimum eigenvalue $\lambda_{\min}$ can go *negative* (not merely close to zero). The SVD-based condition number $|\kappa|$ remains moderate ($\sim$ 7–30), but the sign structure of the matrix is fundamentally wrong for the CG solver. This should be considered a potential concern: any event configuration that triggers indefiniteness will produce unreliable CG results, regardless of the true condition number.

11 Impact on Step 2 Separation Metric

The separation quality is defined as the pooled Cohen's d:

$$d = \frac{\langle x_{\text{true}} \rangle - \langle x_{\text{false}} \rangle}{\sqrt{\frac{1}{2}\left(\sigma^2_{x_{\text{true}}} + \sigma^2_{x_{\text{false}}}\right)}}. \tag{20}$$

Under normal operation: $\langle x_t \rangle = 1.091$, $\langle x_f \rangle = 0.400$, $\sigma_t \approx 0.03$, $\sigma_f \approx 0$, giving $d \approx (0.691/0.021) \approx 33/\sigma_f \to \infty$. In practice with small non-zero σ_f, the computed value is $\approx 5.4\,\sigma$.

Under the anomalous condition at $n = 75$, ± 0.04 rad: $\langle x_t \rangle = 0.662$, $\langle x_f \rangle = 0.393$, $\sigma_t = 6.04$ (inflated by diverged-event outliers), $\sigma_f = 0.487$ (also inflated by the redistributed residual error), giving:

$$d = \frac{0.662 - 0.393}{\sqrt{\frac{1}{2}(6.04^2 + 0.487^2)}} = \frac{0.269}{\sqrt{18.36}} = \frac{0.269}{4.28} \approx 0.06\,\sigma.$$

The separation metric is therefore dominated by the inflated variance caused by the diverged event, not by a genuine loss of Hamiltonian discrimination power.

12 Conclusions

1. The dip in recovery rate and separation quality observed at ± 0.04 rad generation angle and 75 tracks is caused by a **near-coincident track degeneracy** in (at least one of) the five random events sampled at that density point.

2. When two tracks land within $\Delta\phi_{\text{eff}} \approx 0.12$ mrad of each other in angular space, their cross-segments fall within the Hamiltonian's acceptance window $\varepsilon = 0.425$ mrad. This introduces spurious off-diagonal couplings that make the system matrix **indefinite** (negative eigenvalues).

3. **The matrix indefiniteness is a potential concern.** The Hamiltonian matrix is assumed to be symmetric positive definite (SPD) for the CG solver. The analysis reveals that this assumption *can be violated* whenever near-coincident tracks inject sufficient cross-segment couplings. This is not merely an ill-conditioning issue: the SVD-based condition number $|\kappa|$ remains moderate ($\sim$7–30), but the matrix has fundamentally wrong sign structure. The widely quoted "exploding condition number" ($\kappa \sim 10^{15}$) was a **measurement artefact** caused by applying a positive-definite formula ($\kappa = \lambda_{\max}/\lambda_{\min}$) to an indefinite matrix.

4. The CG solver has **no convergence guarantee** on indefinite systems, producing pathological activation values ($[-75, +154]$ observed), which in turn drive the mean activation and separation statistics far from their expected values. A direct solver (LU decomposition) always produces the correct mathematical solution, though the physical interpretation of that solution for an indefinite Hamiltonian remains an open question.

5. The anomaly is **not a systematic failure** of the Hamiltonian design at *typical* track separations. It is a stochastic artefact of the small sample size ($N_{\text{rep}} = 5$) at a generation angle (± 0.04 rad) that packs tracks $25\times$ more densely in angular space than the ± 0.2 rad setting. However, the underlying indefiniteness mechanism will affect any sufficiently dense environment.

6. The ± 0.2 rad and ± 0.1 rad settings are effectively immune to this pathology within a 5-event sample, owing to their lower angular-space track density. At higher multiplicities or in real detector conditions with additional noise, the probability of near-coincident configurations increases and may need to be addressed.

7. The parameter γ affects the severity of the indefiniteness: the old code used $\gamma = 2.0$ (diagonal $= 3.0$) while the current code uses $\gamma = 1.5$ (diagonal $= 2.5$). Higher γ reduces but does not eliminate the problem.

Recommendations

- **Investigate the indefiniteness.** The fact that the Hamiltonian matrix can become indefinite may warrant deeper theoretical investigation. If the matrix is intended to be SPD by construction, then the current coupling structure may need modification (e.g. ensuring $\gamma + \delta$ always exceeds the maximum possible row sum of off-diagonal magnitudes).
- **Use a direct sparse solver** (e.g. `scipy.sparse.linalg.spsolve`) instead of CG for moderate matrix sizes ($N_{\mathrm{seg}} \lesssim 10^5$). Direct solvers produce correct solutions regardless of matrix definiteness.
- **Increase $N_{\mathbf{rep}}$** (e.g. to 20–50 events per density point) to average out single-event anomalies and resolve whether the dip is a genuine trend or a statistical fluctuation.
- **Add a near-coincidence pre-check**: before constructing the Hamiltonian, flag and report pairs of tracks with $\Delta\phi_{\mathrm{eff}} < 2\varepsilon/f_{\mathrm{amp}} \approx 0.25\,\mathrm{mrad}$. Events with flagged pairs can either be excluded or handled specially.
- **Regularisation**: adding a small $\lambda\mathbf{I}$ term to $\mathbf{A}$ (Tikhonov regularisation) would ensure positive definiteness and stabilise CG, but may mask the underlying indefiniteness issue.
- **Re-examine γ**: consider whether $\gamma = 2.0$ or higher provides sufficient diagonal dominance to guarantee positive definiteness for the expected range of track configurations.

A Numerical Data Tables

Step 2 activation statistics for all angle settings

Key segment counts at $n = 75$ tracks

B Step 3 Competition Data at ± 0.04 rad

Table 6: Full Step 2 statistics for all three generation angle settings. Columns: $\langle x_t \rangle$, σ_{x_t}, $\langle x_f \rangle$, recovery (%), separation (σ).

Tracks	$\langle x_t \rangle$	σ_{x_t}	$\langle x_f \rangle$	Recovery (%)	Sep. (σ)
Angle ± 0.2 *rad*					
5	1.085	0.000	0.400	100.0	5.30
10	1.078	0.162	0.400	99.0	5.12
20	1.085	0.000	0.400	100.0	5.30
30	1.073	0.185	0.400	97.2	4.93
50	1.081	0.113	0.400	99.8	5.22
75	1.081	0.146	0.400	100.0	5.24
100	1.083	0.132	0.400	100.0	5.27
150	1.077	0.168	0.400	99.1	5.12
Angle ± 0.1 *rad*					
5–50	1.0909	0.031	0.400	100.0	5.37
75	1.143	0.284	0.400	100.0	3.69
100	1.0909	0.031	0.400	100.0	5.37
150	1.0909	0.031	0.400	100.0	5.37
Angle ± 0.04 *rad*					
5–50	1.0909	0.031	0.400	100.0	5.37
75	**0.662**	**6.04**	**0.393**	**82.7**	**0.06**
100	1.0909	0.031	0.400	100.0	5.37
150	1.0919	0.033	0.400	100.0	5.32

Table 7: Segment structure at $n = 75$ tracks. The total segment count is identical for all angle settings; only the *coupling* (off-diagonal terms) differs.

	Total segments	True segments	False segments
Formula	$4n^2 = 22{,}500$	$4n = 300$	$4n(n-1) = 22{,}200$
Numerical	22,500	300	22,200

Table 8: Step 3 per-hit competition summary for ± 0.04 rad. All values are from a *fresh* set of 5 events, independent of Step 2. The 75-track row (bold) is healthy, confirming the Step 2 anomaly is a sampling artefact.

Tracks	$\langle x_{\text{true}} \rangle$	$\langle \sum x_{\text{false}} \rangle$	$\langle N_{\text{competitors}} \rangle$
5	1.091	2.80	7.0
10	1.091	6.30	15.8
20	1.091	13.30	33.2
30	1.091	20.30	50.8
50	1.091	34.30	85.8
75	**1.125**	**51.84**	**129.5**
100	1.091	69.31	173.2
150	1.079	104.31	260.8